



COURSE SYLLABUS

CS419 – Introduction to Information Retrieval

1. GENERAL INFORMATION

Course name:	Introduction to Information Retrieval
Course name (in Vietnamese):	Nhập môn truy xuất thông tin
Course ID:	CS419
Knowledge block:	Computer Science
Number of credits:	4
Credit hours for theory:	40
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	N/A
Prior-course:	N/A
Instructors:	Dr. Nguyễn Vinh Tiệp Dr. Lê Trung Nghĩa

2. COURSE DESCRIPTION

The course is designed to provide students skills to understand and develop information retrieval systems. Through lectures and practical in-class exercises aiming at developing search systems, students are exposed to relevant methods, processes, and techniques for:

- Reading and analyzing English articles,
- Developing search algorithms,
- Designing interfaces for search systems.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Understand and describe text retrieval algorithms	

G2	Understand and describe image retrieval algorithms	
G3	Understand and describe search systems	
G4	Ability to self-learn and update knowledge	
G5	Effective teamwork and presentation	

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Understand definitions	I/T
G1.2	Understand evaluation measures	I/T
G1.3	Understand text retrieval algorithms	I/T
G2.1	Understand and describe visual features	I/T
G2.2	Understand image retrieval algorithms	I/T
G3.1	Understand interactive search systems	I/T
G3.2	Design interfaces for search systems	U
G4.1	Read English materials	U
G4.2	Self-learn and update knowledge	U
G5.1	Develop effective personal communication skills	U
G5.2	Improve critical thinking skills	U

5. TEACHING PLAN

THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	S01: Course Introduction S02: Web Crawling	G1.1, G1.3, G4.1, G4.2	Lecturing Q&A, Group discussion	HW1
2	S03: Boolean Retrieval S04: Bag of Words	G1.1, G1.3, G4.1, G4.2	Lecturing Demonstration, Q&A	HW2

3	S05: Local Features S06: Bag of Visual Features	G1.1, G2.1, G2.2, G4.1, G4.2	Lecturing Demonstration, Q&A	HW3
4	S07: Indexing Construction S08: Evaluation of IR Systems	G1.1, G1.2, G1.3, G4.1, G4.2	Lecturing Q&A, Group discussion	
5	S09: Text Retrieval Systems	G1.1, G1.3, G4.1, G4.2	Lecturing Demonstration, Q&A	
6	S10: Image Retrieval Systems	G1.1, G2.1, G2.2, G4.1, G4.2	Lecturing Demonstration, Q&A	
7	S11: Image Encoding	G1.1, G2.1, G2.2, G4.1, G4.2	Lecturing Demonstration, Q&A	HW4
8	<i>S12: Interface Design</i>	G3.2, G4.1, G4.2	<i>Tutoring, Discussion</i>	<i>HW5</i>
9	S13: Query Operations S14: Interactive Search	G1.1, G3.1, G4.1, G4.2	Lecturing Q&A, Group discussion	
10	S15: Review	All Gs	Lecturing Q&A, Discussion	

LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Text search systems	G1.1, G4.1, G4.2, G5.1, G5.2	Demonstration, Q&A	LW1

2	Image search systems	G1.1, G4.1, G4.2, G5.1, G5.2	Demonstration, Q&A	LW2
3	Interface design	G4.1, G4.2, G5.1, G5.2	Demonstration, Q&A	LW3
4	Interactive search	G1.1, G4.1, G4.2, G5.1, G5.2	Demonstration, Q&A	LW4

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			40%
A11	Group assignments (GA):	GUI programming	All Gs	20%
A12	Personal assignments (PA):	Analyze and develop search algorithms	All Gs	20%
A2	Exams			60%
A21	Project	Design search systems on computers	All Gs	40%
A31	Final exam	Q&A examination about designed systems	All Gs	20%

7. RESOURCES

Textbooks

- [1] C. Manning, P. Raghavan, H. Schütze, Introduction to Information Retrieval, 2008.
- [2] Ricardo Baeza-Yates and Berthier Ribeiro-Neto, Addison-Wesley, Modern Information Retrieval, 2011
- [3] Stefan Buttcher, Charlie Clarke, Gordon Cormack, Information Retrieval: Implementing and Evaluating Search Engines, MIT Press, 2010.

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students are encouraged to form study groups to exchange their ideas. However, individual work must be done and submitted on your own.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students who are absent for more than 3 theory sessions WITHOUT permission are not allowed to take the exams.
- Students can communicate with lecturers and teaching assistants by writing posts on Moodle discussion forum or sending messages to their faculty email addresses.